PAK-AUSTRIA FACHHOCHSCHULE:

INSTITUTE OF APPLIED SCIENCES AND TECHNOLOGY

Machine Learning Lab – Final Project Report

Multiclass Heart Disease Classification Using Machine Learning

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1. Introduction

Heart disease is one of the leading causes of mortality worldwide and remains a major public health concern. Early identification and correct classification of heart disease types play a crucial role in timely treatment and prevention of severe complications. With the growing availability of medical data, machine learning techniques can be effectively used to identify hidden patterns in patient records and assist healthcare professionals in decision-making.

The objective of this project is to design and implement a **multiclass machine learning classification system** that predicts the **type of heart disease** based on patient demographics, medical history, symptoms, and laboratory measurements. The project follows a complete machine learning workflow, including dataset selection, preprocessing, exploratory data analysis (EDA), model development, evaluation, final model selection, and deployment through a Streamlit-based interface.

This project strictly follows the **Machine Learning Lab guidelines**, using **classical machine learning models only**, performing meaningful preprocessing on a real-world dataset, and integrating the trained model with a simple user interface .

2. Dataset Description

2.1 Dataset Source

The dataset used in this project was originally provided as a **test dataset during the mid-term examination** of the Machine Learning Lab course. The dataset was designed to replicate **real-world medical data** and was collected by a group of undergraduate students as part of their **Final Year Project (FYP)**. As a result, the dataset contains realistic characteristics such as missing values, inconsistent labels, noisy categorical features, and non-uniform data distributions.
During the mid-term examination, I faced challenges while working with this dataset due to its unclean structure and limited size. In order to better understand the data and improve practical machine learning skills, the same dataset was revisited for the final project and analyzed in greater depth.
Initially, the dataset contained **approximately 300 records**, which did not meet the minimum dataset size requirement specified in the lab guidelines. Therefore, the dataset was **expanded to 500 records** by adding 200 additional samples based on the original data distribution, while ensuring that no artificial class imbalance was introduced. This step allowed the dataset to meet the course requirements while preserving its original characteristics.

This dataset was selected intentionally to simulate the challenges encountered in real-world medical data analysis and to demonstrate the application of proper preprocessing, exploratory data analysis, and model evaluation techniques.

2.2 Dataset Overview

- **Number of records:** 500
- **Number of features:** 29
- **Target variable:** `Prediction` (Type of Heart Disease)

The dataset contains a mixture of numerical, binary, and categorical medical features related to patient health conditions, lifestyle factors, symptoms, and laboratory test results. Due to missing values, categorical inconsistencies, and noisy medical attributes, the dataset requires real preprocessing, making it suitable for a machine learning lab project as per the guidelines .

2.3 Feature Description

Below are some of the key features included in the dataset:

- **Gender:** Biological sex of the patient
- **Age:** Age of the patient in years
- **Hypertension:** Indicates whether the patient has high blood pressure
- **Heart Disease:** Past history of heart-related conditions
- **Family History:** Presence of heart disease in close family members
- **Chest Pain:** Indicator of chest pain symptoms
- **Max Heart Rate:** Maximum heart rate achieved during activity
- **Average Glucose Level:** Blood glucose measurement
- **Cholesterol:** Serum cholesterol level
- **Smoking Status:** Smoking habits of the patient
- **Serum Creatinine:** Indicator of kidney function
- **Serum Sodium:** Sodium concentration in blood
- **Body Part:** Body region where pain or discomfort is experienced

The **target variable (`Prediction`)** contains the following classes:

- Normal
- Heart Failure
- Arrhythmia
- Coronary Artery Disease (CAD)
- Cardiomyopathy
- Congenital Heart Disease
- Unknown

The presence of multiple disease categories makes this a **challenging multiclass classification problem**, which aligns well with the objectives of the Machine Learning Lab.

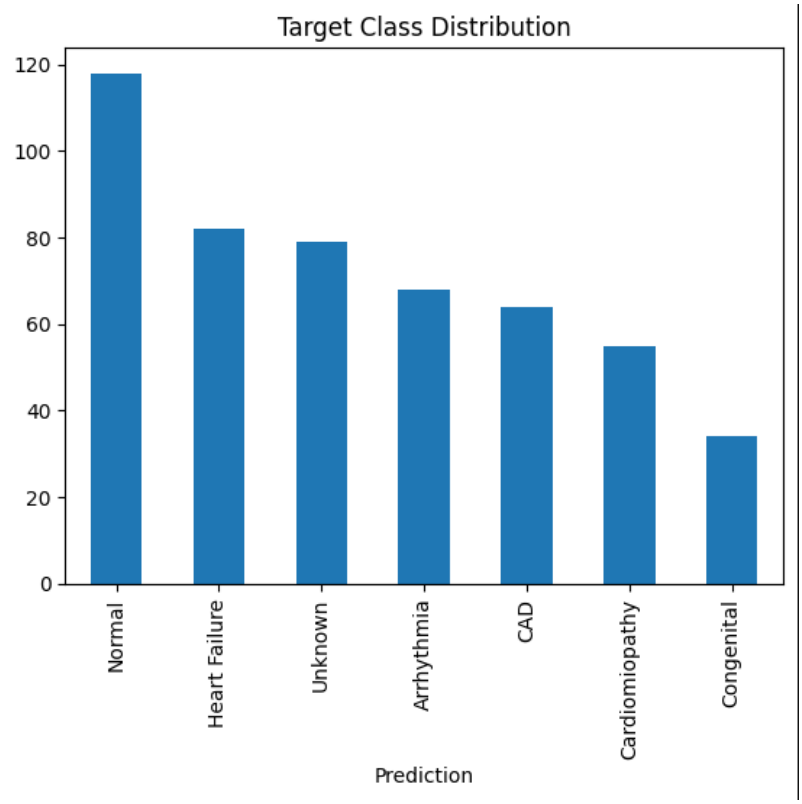
3. Data Preprocessing

Preprocessing was a critical step in this project, as the dataset contained missing values, inconsistent labels, categorical variables, and outliers. All preprocessing steps were implemented carefully to avoid data leakage, as required by the lab guidelines .

3.1 Target Label Normalization

The target variable contained inconsistencies due to casing and spelling variations (e.g., *Normal*, *normal*, *NORMAL*, *Heart failure*, *heart Failure*). These inconsistencies can mislead machine learning models by treating identical medical conditions as separate classes.

To address this issue, target labels were normalized to ensure semantic consistency **without altering the underlying class distribution**.



3.2 Handling Missing Values

- The **Swelling** feature contained a large number of missing values and was therefore removed to prevent biased imputation.
- **Numerical missing values** were handled using **median imputation**, which is robust to outliers.
- **Categorical missing values** were handled using **most frequent (mode) imputation**.
- All imputation was performed **inside the preprocessing pipeline** to prevent data leakage between training and testing sets.

3.3 Handling Categorical Variables

Categorical features such as Gender, Smoking Status, Family History, and symptom-related variables were encoded using **One-Hot Encoding**. This approach allows machine learning models to process categorical information without introducing ordinal bias.

3.4 Feature Scaling

Numerical features were standardized using **StandardScaler**. Feature scaling was necessary to ensure that features with larger numeric ranges (e.g., cholesterol, glucose levels) did not dominate model learning, particularly for linear models such as Logistic Regression.

3.5 Outlier Detection and Handling

Exploratory analysis revealed the presence of outliers in features such as cholesterol. These values were **not removed**, as extreme medical values may represent valid high-risk patients rather than data errors. Instead, robust tree-based models were selected to handle such values naturally.

3.6 Train–Test Split

The dataset was split into:

- **80% training data**
- **20% testing data**

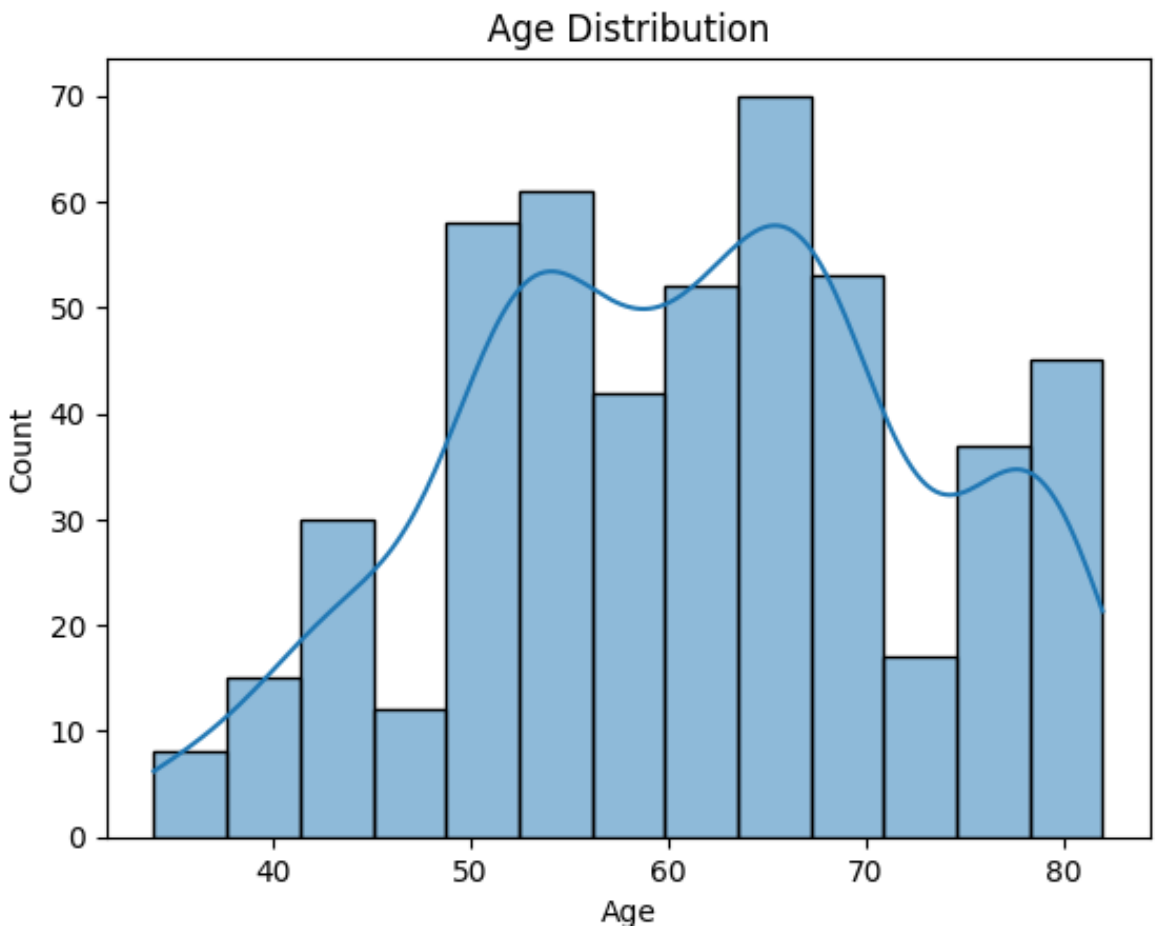
A **stratified split** was used to preserve the class distribution across both sets.

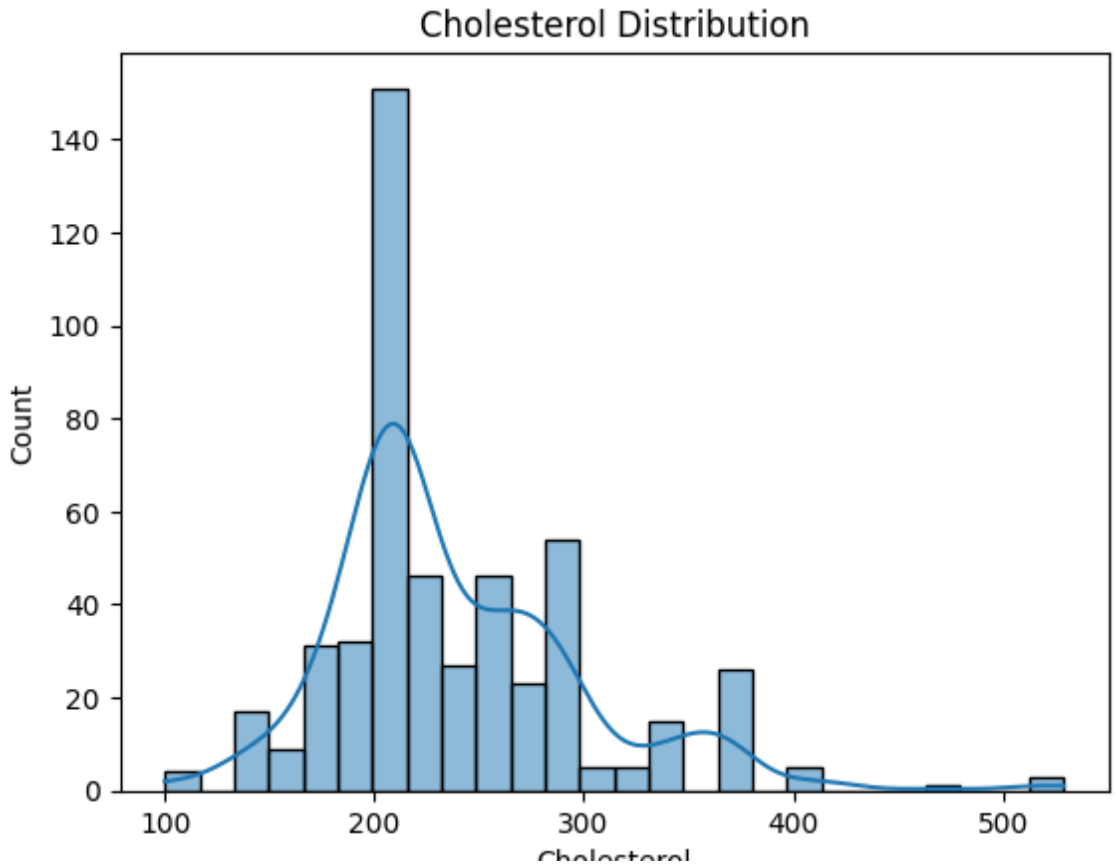
4. Exploratory Data Analysis (EDA)

EDA was performed to understand feature distributions, relationships, and patterns in the data, as required by the lab guidelines .

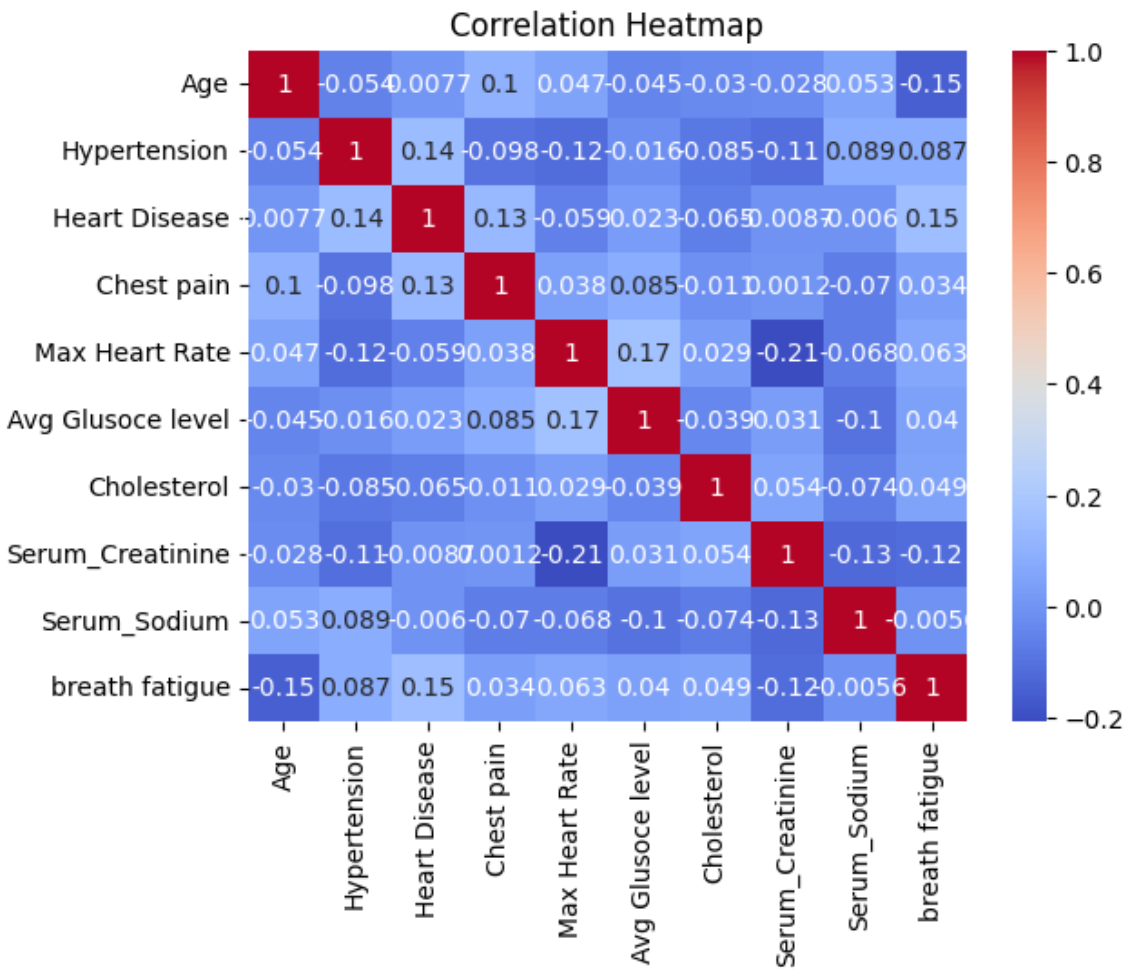
4.1 Feature Distributions

The age distribution shows that most patients fall within the middle-aged to elderly range, which aligns with real-world heart disease trends. Cholesterol levels exhibit a right-skewed distribution, indicating the presence of high-risk patients with elevated cholesterol values.




Cholesterol Distribution
Count
0
20
40
60
80
100
120
140
100
200
300
400
500

4.2 Correlation Analysis



The correlation matrix shows mostly weak to moderate correlations between numerical features, suggesting low multicollinearity and supporting the use of ensemble models capable of learning non-linear relationships.

4.3 Summary Statistics

Summary statistics further confirm the presence of variability in medical attributes, reinforcing the need for preprocessing and robust model selection.

5. Model Development

The following machine learning models were trained in accordance with the lab requirement of implementing **at least two ML models** :

1. **Logistic Regression**
2. **Random Forest Classifier**
3. **Gradient Boosting Classifier**

Each model was trained using the same preprocessing pipeline to ensure a fair comparison.

6. Model Evaluation

6.1 Evaluation Metrics

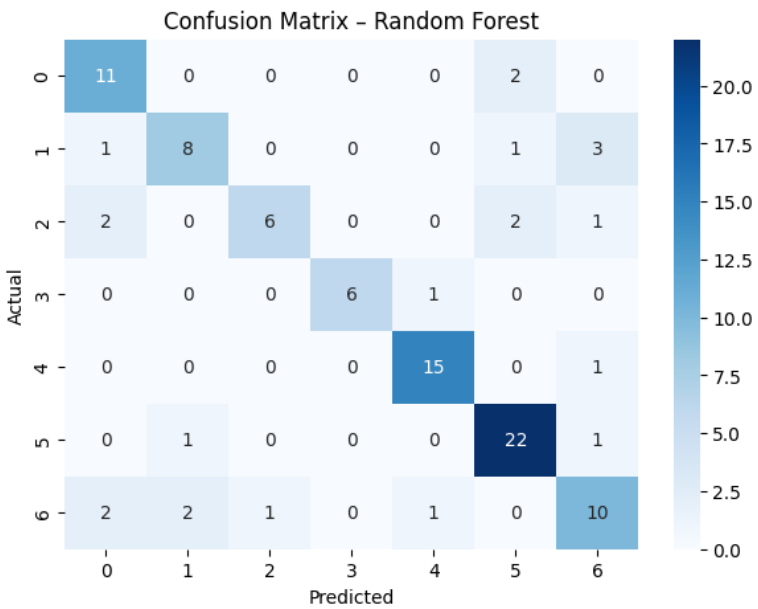
The models were evaluated using:

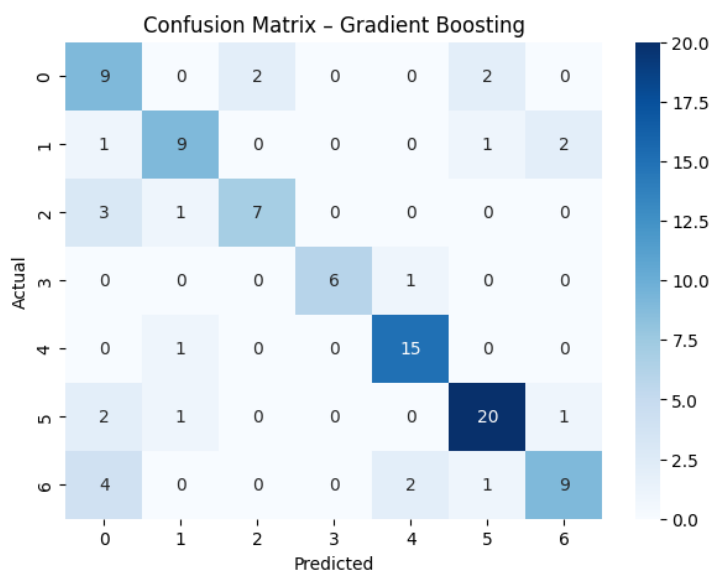
- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix
- ROC Curves (One-vs-Rest for multiclass classification)

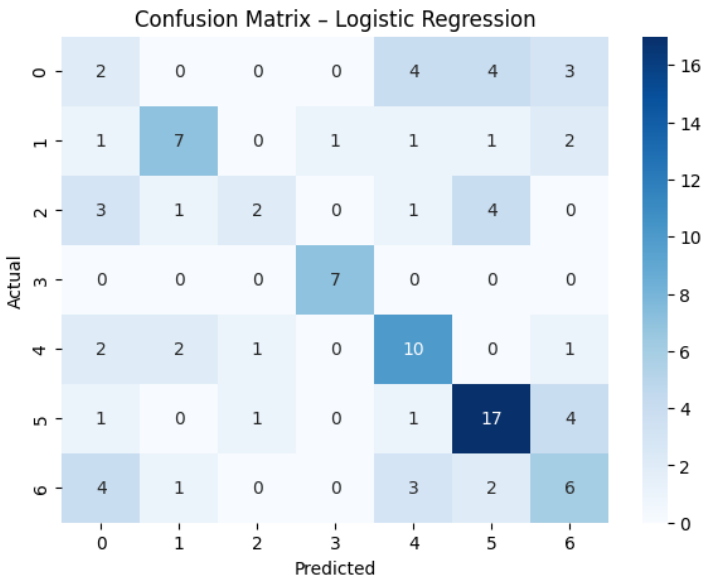
	Model	Accuracy	Precision (Macro)	Recall (Macro)	F1-score (Macro)
0	Random Forest	0.78	0.799155	0.763329	0.773124
1	Gradient Boosting	0.75	0.774018	0.744494	0.752017
2	Logistic Regression	0.51	0.521050	0.511780	0.503083

Table shows the performance of all trained models. Random Forest achieved the highest accuracy and macro F1-score, indicating balanced performance across all heart disease classes. Logistic Regression performed poorly due to its linear nature, while Gradient Boosting showed competitive but slightly lower performance compared to Random Forest.

6.2 Confusion Matrices

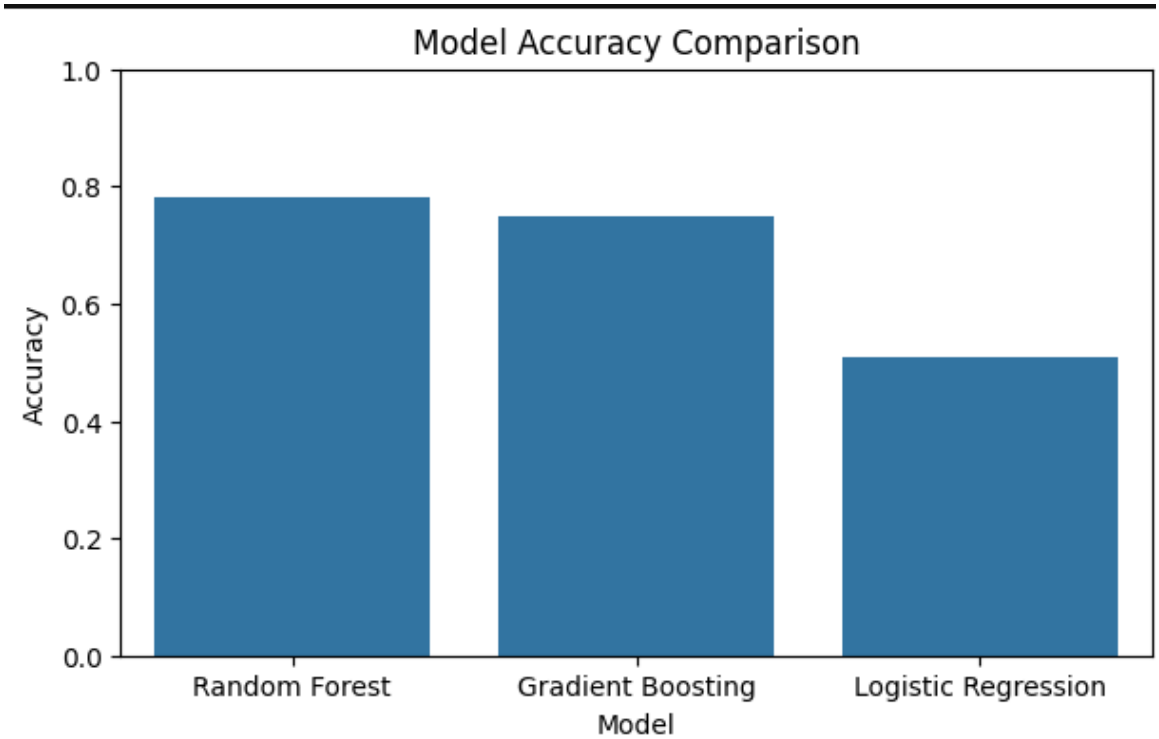






The confusion matrices show that Random Forest performs consistently well across most classes, while Logistic Regression struggles with complex class boundaries.

6.3 Model Comparison



Random Forest achieved the highest accuracy, followed by Gradient Boosting, while Logistic Regression showed comparatively lower performance.

7. Final Model Selection

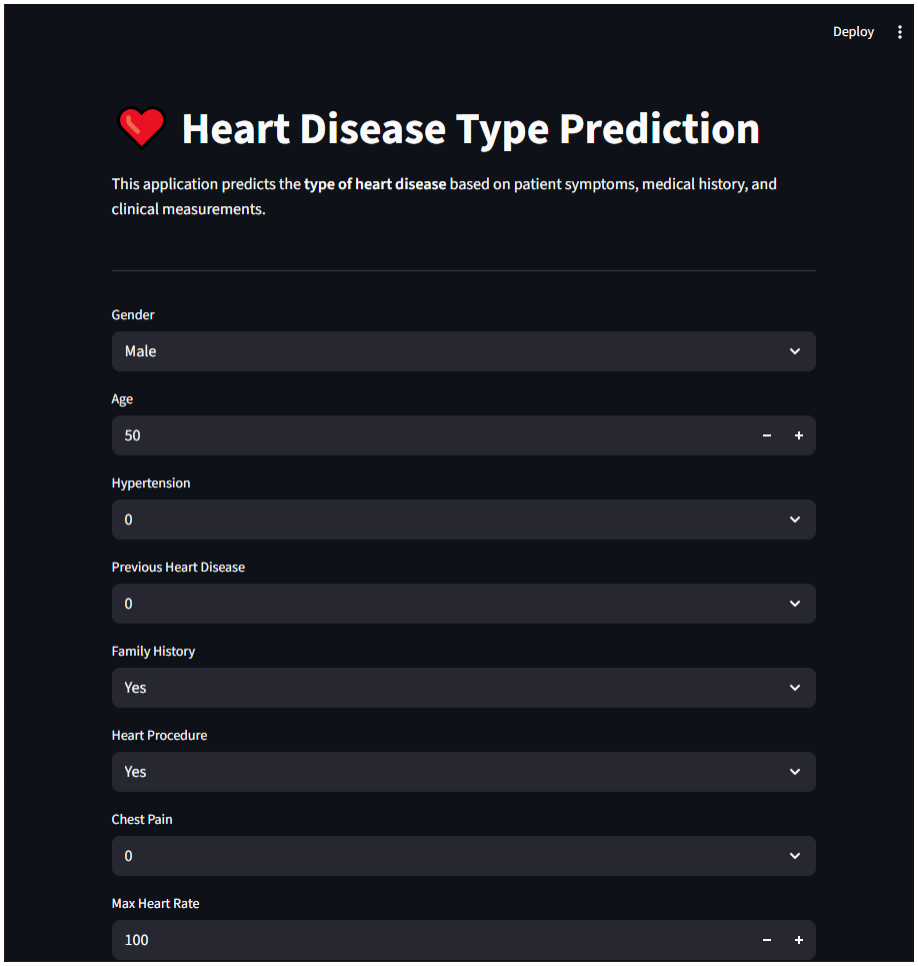
Random Forest was selected as the final model due to its:

- Superior accuracy
- Robustness to noisy medical data
- Ability to model complex non-linear relationships

8. Interface Development

A simple **Streamlit-based interface** was developed that:

- Allows users to input patient medical information
- Displays the predicted type of heart disease



9. Conclusion

This project successfully demonstrates the complete application of machine learning techniques for multiclass heart disease classification. From preprocessing a noisy medical dataset to training and evaluating multiple models and deploying a user interface, all stages of the ML workflow were implemented in accordance with the Machine Learning Lab guidelines .